SE444: Introduction to Artificial Intelligence
Makeup Mid-Term II Fall 2025

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Date: Wednesday, December 10, 2025 Time: 90 minutes

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Clearly state any assumptions made that might be needed to understand your solution.
- Show your work – any sign of serious effort will be considered.
- No marks will be given for not making any effort.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	Mapped CLOs
Part I MCQ	FOL — Chained Implications	5 min	1		CLO 1
	Propositional Logic — Satisfiability	5 min	1		CLO 1
	CSP — Arc Consistency	5 min	1		CLO 1, 2
	Logical Agents — Environment Types	5 min	1		CLO 1, 3
Part II FOL	Translation to FOL	10 min	3		CLO 1
	Translation with Negation	10 min	3		CLO 1
	Negation and CNF Conversion	20 min	4		CLO 1
Part III CSP	Formulate Conference Scheduling as CSP	10 min	2		CLO 2
	Apply Arc Consistency (AC-3)	15 min	3		CLO 2
	Find a Consistent Solution	5 min	1		CLO 2
Total for Part I (MCQ)		20 min	4		
Total for Part II (FOL)		40 min	10		
Total for Part III (CSP)		30 min	6		
Total Points		90 min	20		

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 4 points **Total Time:** 20 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Question I.1: First-Order Logic (FOL) — Chained Implications

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

You are given the following FOL sentences (assume a non-empty domain):

1. $\forall x\,(R(x) \rightarrow S(x))$
2. $\forall x\,(S(x) \rightarrow T(x))$
3. $\exists x\,R(x)$

Which of the following sentences are logically entailed by (1), (2), and (3)?

A) $\exists x\,T(x)$

B) $\forall x\,T(x)$

C) $\exists x\,(R(x) \wedge T(x))$

D) $\forall x\,(T(x) \rightarrow R(x))$

E) $\exists x\,(S(x) \wedge \neg R(x))$

Question I.2: Propositional Logic — Satisfiability Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Consider a knowledge base (KB) with propositional atoms a, b, c, d, e:

1. $(a \wedge b) \rightarrow c$
2. $c \rightarrow d$
3. $d \rightarrow e$
4. $\neg e$
5. b

Which of the following statements is true?

A) KB is unsatisfiable.

B) KB $\models a$.

C) KB $\models \neg a$.

D) KB $\models c$.

E) KB can be satisfied with $a = \text{true}$ and $c = \text{true}$.

Question I.3: CSP — Arc Consistency (AC)

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 2

Consider a CSP with variables X, Y, Z.
Initial Domains: $D(X) = \{1, 2, 3\}$, $D(Y) = \{2\}$, $D(Z) = \{1, 2, 3\}$
Constraints:

- $X \neq Y$
- $Y < Z$
- $X < Z$

Run AC-3 (Arc Consistency) checks and remove impossible values until nothing changes.

Which are true?

A) After AC, $D(Z)$ becomes $\{3\}$.

B) After AC, $D(X)$ becomes $\{1\}$.

C) After AC, all domains become singletons and the CSP is solved without search.

D) After AC, $D(X)$ still contains the value 2.

E) If we change the constraint "$Y < Z$" to "$Y \leq Z$", then $D(Z)$ would include $\{2, 3\}$ after AC.

Question I.4: Logical (Knowledge-Based) Agents — Environment Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 3

A hospital deploys a **medical diagnosis robot** operating in a complex healthcare environment that is **partially observable** (limited access to complete patient history), **stochastic** (uncertain test results and treatment responses), **sequential** (each diagnosis affects treatment decisions), and **dynamic** (patient conditions change over time).

Evaluate the following statements about possible agent architectures and identify which are correct.

A) A simple reflex agent mapping current symptoms directly to diagnoses would be sufficient for this environment if the symptom-to-diagnosis mapping is comprehensive enough.

B) A model-based reflex agent could track a patient's medical history and previous test results in an internal state, enabling more accurate diagnoses than a memoryless agent.

C) A goal-based agent cannot operate rationally in this environment because treatment outcomes are uncertain, making goal achievement unpredictable.

D) A utility-based agent could weigh the expected benefits and risks of different diagnostic tests, potentially ordering additional tests when the value of information justifies the cost.

E) A learning agent would not provide any advantage over a pre-programmed agent in this domain since medical knowledge is fixed and does not change.

Part II. Logical Reasoning in Healthcare Context

Total Score: 10 points **Total Time:** 40 minutes

Given Predicates:

Predicate	Meaning
Doctor(x)	x is a doctor
Patient(x)	x is a patient
Hospital(x)	x is a hospital
WorksAt(d, h)	Doctor d works at hospital h
TreatedBy(p, d)	Patient p is treated by doctor d
HasDisease(p, dis)	Patient p has disease dis
Disease(dis)	dis is a disease
Prescribes(d, p, m)	Doctor d prescribes medication m to patient p
Medication(m)	m is a medication
Treats(m, dis)	Medication m treats disease dis
Examined(d, p)	Doctor d has examined patient p

Translate each of the following statements into first-order logic (FOL) using the provided predicates, then perform the required logical transformation.

Question II.1: Translation to FOL

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):

"Every patient who is treated by a doctor at a hospital is prescribed at least one medication that treats one of their diseases."

Question II.2: Translation with Negation

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):

"No doctor prescribes medication to a patient they have not examined."

Question II.3: Negation and Conjunctive Normal Form (CNF)

Time: 20 minutes **Score:** 4 points **Mapped CLOs:** CLO 1

Negate the following statement and convert it step-by-step into Conjunctive Normal Form (CNF). Show all transformations clearly and justify each step using the appropriate logical rule (e.g., $\neg\forall$, implication removal, De Morgan's law, or Skolemization).

$$\forall x \left(\text{Doctor}(x) \rightarrow \exists y \left(\text{Patient}(y) \land \text{TreatedBy}(y, x)\right)\right)$$

Part III. Constraint Satisfaction Problem (CSP)

Total Score: 6 points **Total Time:** 30 minutes

Part III: Constraint Satisfaction Problems (CSP)

Context: You are organizing the schedule for **AI Conference presentation sessions** on Wednesday. There are three presentation rooms and three keynote speakers available. Each speaker can present in at most one room at a time, and some sessions overlap in time. Your goal is to assign each room to a speaker in a way that satisfies all constraints.

Rooms and Session Times

Room	Topic	Time
Room 1	AI Ethics	10:00–11:00 am
Room 2	Machine Learning	10:30–11:30 am
Room 3	Robotics	11:00–12:00 pm

Speaker Availability

Speaker	Available for Rooms
Dr. Smith	Rooms 1 and 2
Dr. Jones	Rooms 2 and 3
Dr. Brown	Rooms 1, 2, and 3

Assignment Rules (Constraints)

1. Each room must have exactly one speaker assigned.
2. A speaker cannot present in overlapping sessions (time conflicts).
3. A speaker can only be assigned to rooms where they are available.

Question III.1: Formulate as a CSP

Time: 10 minutes **Score:** 2 points **Mapped CLOs:** CLO 2

You are asked to formulate the problem of the conference scheduling scenario as a **Constraint Satisfaction Problem (CSP)**.

Clearly specify:

- the **variables**,
- the **domains** of each variable, and

- the **binary constraints** that must hold between the variables.

You may describe the constraints in English words.

Question III.2: Apply Arc Consistency (AC-3)

Time: 15 minutes **Score:** 3 points **Mapped CLOs:** CLO 2

Using the CSP formulation from Question III.1, apply the **Arc Consistency (AC-3)** algorithm.

- Check each relevant arc between room variables based on the binary constraints.
- For each value in a variable's domain, determine whether it has at least one supporting value in the connected variable's domain.
- Indicate which (if any) domain values are removed and explain why.
- Continue until a **fixed point** is reached (no further pruning possible).
- State the **final domains** for all variables after enforcing arc consistency.

Hint: For constraints of the form $R_i \neq R_j$, a value x in $D(R_i)$ is supported if there exists at least one y in $D(R_j)$ such that $x \neq y$.

Question III.3: Find a Consistent Solution

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 2

Based on your AC-3 results from Question III.2, provide **one valid assignment** that satisfies all constraints and briefly explain why this assignment satisfies all constraints.